

SIMRAN SINGH

Computer Science Undergraduate | Aspiring Backend / Java Developer
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OBJECTIVE

Computer Science undergraduate with a strong foundation in Java and Python development, data structures & algorithms, and database integration. Currently pursuing Advanced Java (J2EE) training covering Servlets and JSP. Seeking a Software Development / Backend Engineering internship to apply hands-on experience building distributed systems, REST APIs, and scalable backend services.

TECHNICAL SKILLS

Languages: Java, Python, Bash

Core CS: Object-Oriented Programming, Data Structures & Algorithms, Operating Systems

Backend & Frameworks: Spring Boot, REST APIs, Servlets & JSP (in progress), Java Swing

Database: MySQL

Tools: Git, GitHub, Docker, Redis, IntelliJ IDEA, VS Code, Apache NetBeans

Problem Solving: 100+ problems solved on LeetCode and 200+ problems on CodeChef, applying pattern-based approaches across arrays, strings, and linked lists

FEATURED PROJECTS

Distributed Rate Limiter — *Java, Spring Boot, Redis, Docker, Lua* [\[GitHub\]](#)

- Built a distributed rate limiter as a Spring Boot REST API implementing two algorithms — Token Bucket (burst-tolerant, HASH-based) and Sliding Window (strict, ZSET-based) — to control request throughput across multiple service instances.
- Eliminated race conditions under concurrent load by executing rate-limit checks as atomic Redis Lua scripts, ensuring consistent enforcement across horizontally scaled instances without locking.
- Designed a fail-open policy so requests are still served if Redis becomes unavailable, prioritizing service availability over strict enforcement.
- Validated correctness and edge cases (burst limits, window resets, concurrent access) with a JUnit 5 + Mockito test suite, and containerized the service with Docker for reproducible local and CI environments.

Bank Management System — *Java, Swing, JDBC, MySQL* [\[GitHub\]](#)

- Built a Java Swing desktop application connected to MySQL via JDBC, supporting 5 core banking operations: account creation, deposits, withdrawals, balance enquiry, and PIN management.
- Applied OOP principles (encapsulation, modularity) across a multi-class architecture to keep the banking workflow maintainable and extensible.
- Structured JDBC connections through a dedicated data-access layer to isolate SQL logic from UI code, reducing coupling and simplifying future extensions.

IP Finder — *Java, Swing, Java Networking (java.net)* [\[GitHub\]](#)

- Developed a Java desktop application using Java's java.net package to retrieve and display the system's host name and IP address via InetAddress.
- Designed a simple Swing GUI so non-technical users can access low-level networking information without using the command line.
- Implemented exception handling for network lookup failures to keep the application stable across different environments.

EDUCATION

KIPM College of Engineering and Technology — Gida, Uttar Pradesh

B.Tech, Computer Science & Engineering | CGPA: 8.45 | 2024 – 2028